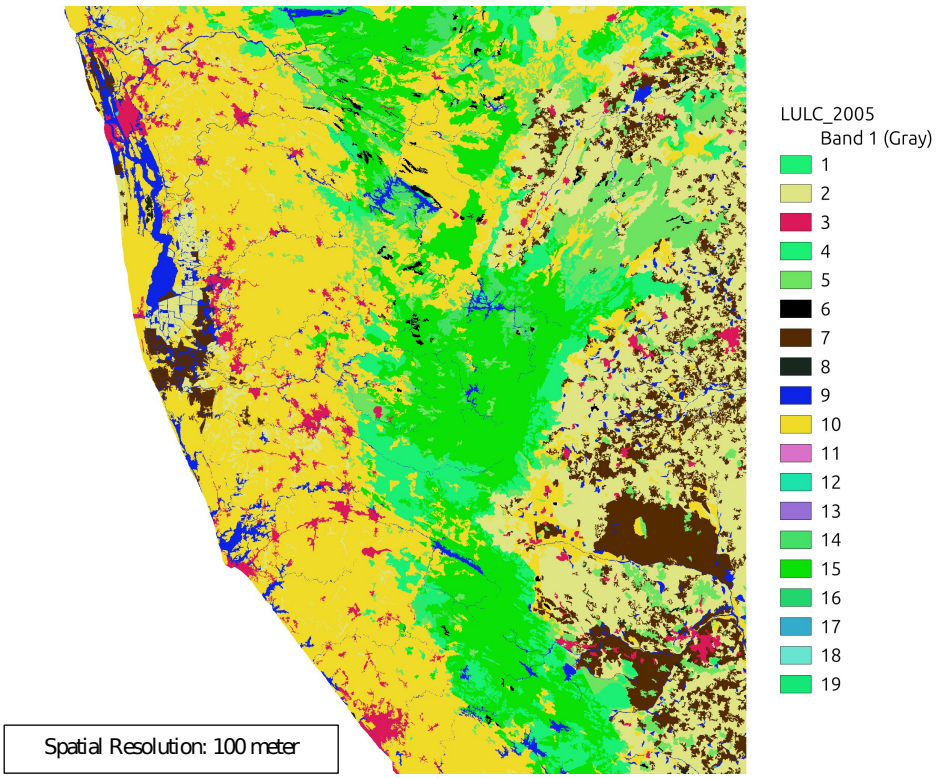
Land Use and Land Cover (LULC) Classification Map

Pixel values / Classification:

1- Deciduous Broadleaf Forest
2- Cropland
3- Built-up Land
4- Mixed Forest
5- Shrubland
6- Barren Land
7- Fallow Land
8- Wasteland
9- Water Bodies
10- Plantations
11- Aquaculture
12- Mangrove Forest
13- Salt Pan
14- Grassland
15- Evergreen Broadleaf Forest
16- Deciduous Needleleaf Forest
17- Permanent Wetlands
18- Snow & Ice
19- Evergreen Needleleaf Forest



Data on Elephant Ecology

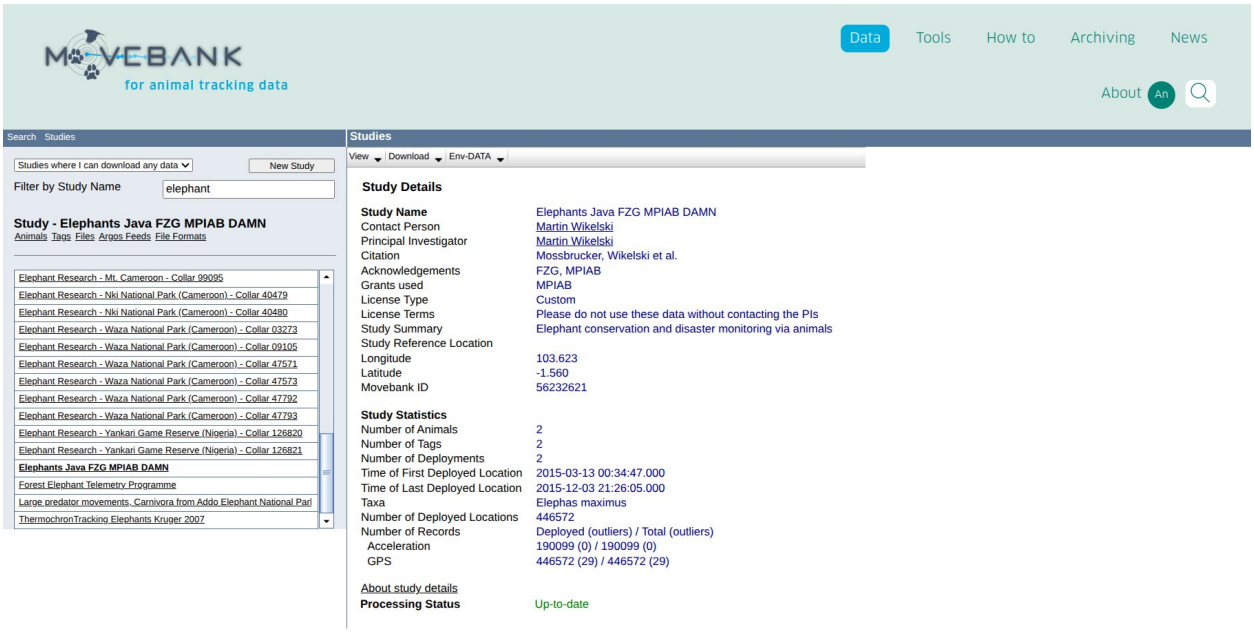
Parameter	Value	Comments	Supporting References
Life Expectancy	60-70 years (maximum) in the wild		
Reproduction: Age at first reproduction in the wild	13–15 yr		
Reproduction: Gestation Period	18 to 22 months (nearly 2 years)		
Reproduction: Calving Interval	Every 4 to 5 yrs	female elephant may give birth to 4 to 7 calves over her entire lifespan	
Social Dynamics: Females	Matriarchal structure and are led by females. Herd consists of 4 to 8 closely related females and their dependent offspring.		
Social Dynamics: Males	Young males leave herds once reach puberty (around 12 to 15 years old), mostly solitary or in temporary associations.		
Movement and Ranging: Home Range Size	500 km² for females (seasonal) 750 km² for males (seasonal)	Males generally range farther than females. Home ranges varies hugely by habitat quality and fragmentation.	
Movement and Ranging: Daily Displacement	5–10 km/day		
Size/Age Class	Calf 0–1 Juvenile 1–5 Sub-adult 5–15 (♀) / 5–20 (♂) Young adult 15–25 (♀) / 20–30 (♂) Adult 25+ (♀) / 30+ (♂)		

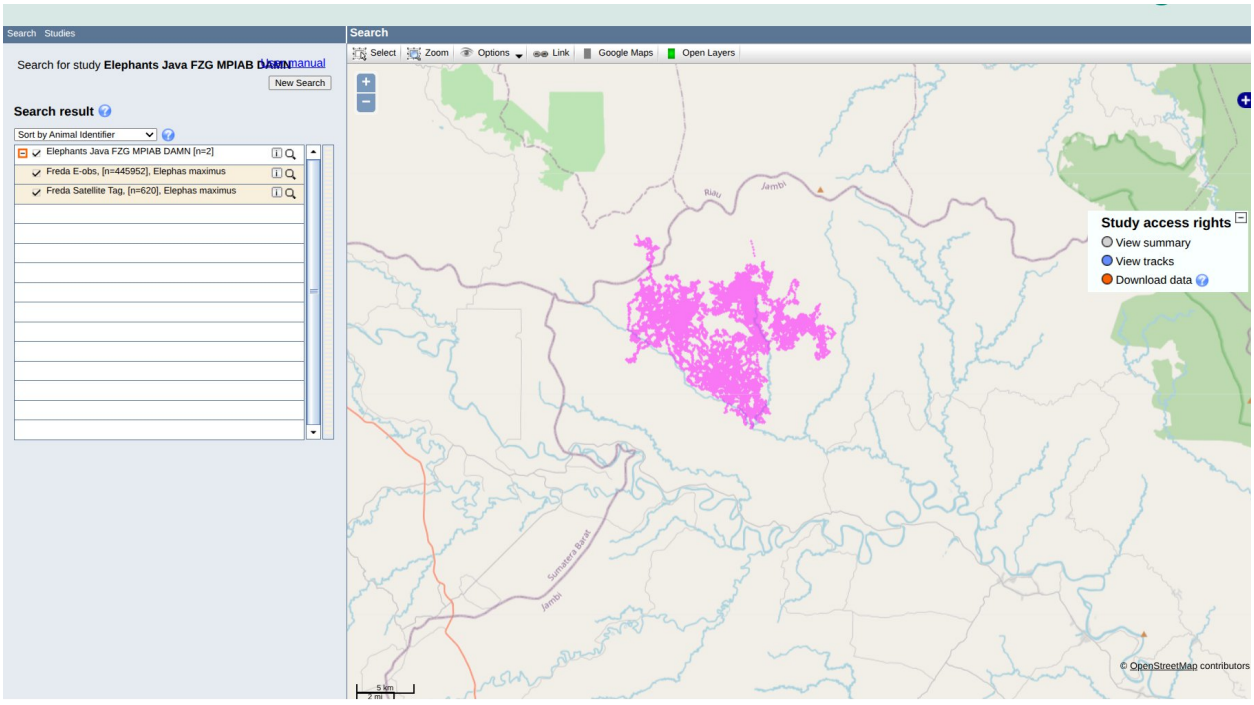
Age-Specific Mortality	Infant/Juvenile Mortality (1-5 yrs): 10-30% Prime-Adult Mortality (10-45 yrs): 2-3% Senescent Mortality (after 45-50 yrs): 30%	Follows a U-shaped curve where death rates are high at the very beginning and very end of life. Male mortality higher than females.	
Anthropogenic (Human-Induced) Mortality	**Poaching mortality** is toward older adults with large tusks. **Crop-raiding** behavior correlates to a higher daily probability of death.		
Factors Affecting Movement	1. **Water availability**: Elephants stay close to rivers/waterholes, especially in dry season (water source visit frequency 1-3 days). 2. **Forage/vegetation quality**: Elephants track resource distribution within the landscape. 3. **Elevation/slope/terrain ruggedness**: Strong preference for low elevation and gentle slopes (0–10°); rugged terrain avoided (energetic costs). 4. **Distance to human settlements**: Avoided during day and used more at night. 5. **Roads & barriers**: physical/psychological barriers, fragmenting ranges and altering paths.		

Movement Datasets

1. Asian elephant (*Elephas maximus*) - Java (Indonesia)

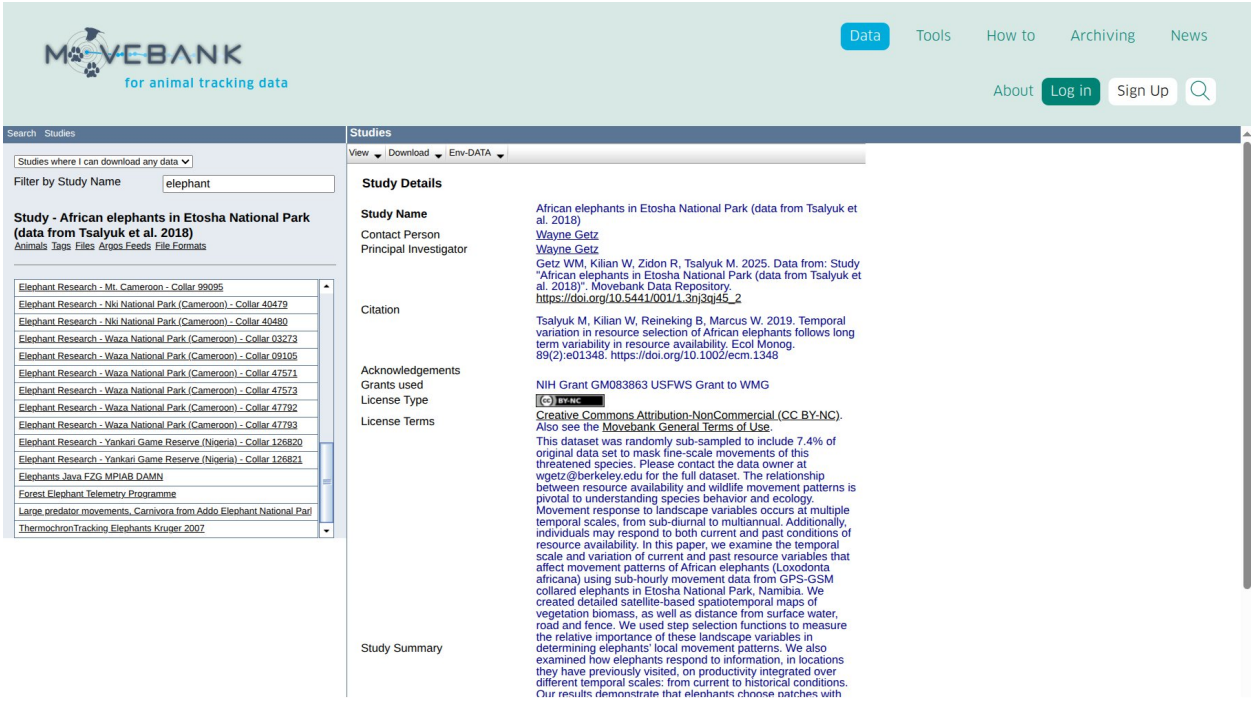
https://www.movebank.org/cms/webapp?gwt_fragment=page=studies,path=study56232621

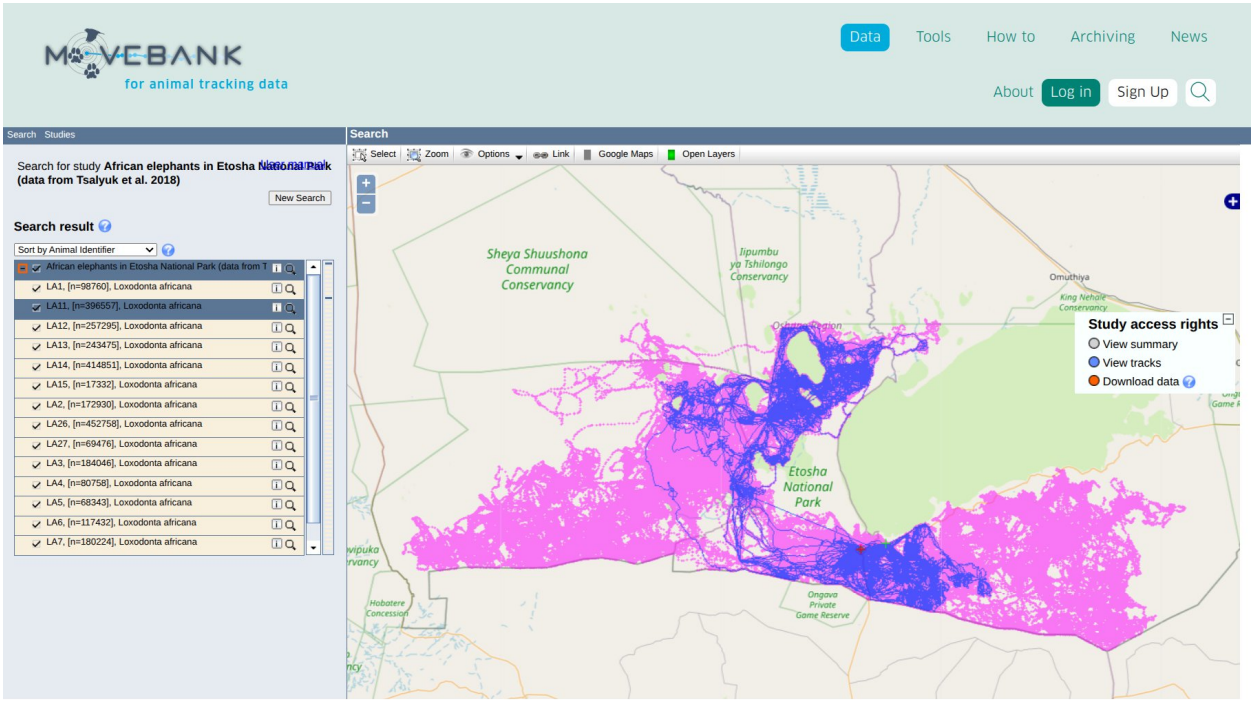




2. African elephants (*Loxodonta africana*) in Etosha National Park (data from Tsalyuk et al. 2018)

https://www.movebank.org/cms/webapp?gwt_fragment=page=studies,path=study605129389





3. African elephants (*Loxodonta africana*) in Kruger National Park (data from Thaker et al. 2019)

https://www.movebank.org/cms/webapp?gwt_fragment=page=studies,path=study736029750

